

Chapter 1

C# Strings, Structures & Collections (with Embedded Review)

ENEL712 Embedded Systems Design — Lecture 20

Summary

C# strings, structures, and collection classes, paired with a recap of foundational embedded concepts (MCU architecture, memory, comms, and software design).

Topics Covered

- C# strings
- C# structures (struct)
- Collection classes
- Embedded review: MCU architecture and operation
- Embedded review: MCU memory types
- Embedded review: digital communications
- Embedded review: software design and estimation

1.1 C# Strings

- String = specialised array of 16-bit UTF-8 characters (backward-compatible with ASCII).
- Concatenate with +.
- Substring(startIndex, count) extracts.
- IndexOf(substring) is case-sensitive; returns -1 if not found.
- Trim/TrimStart/TrimEnd manage whitespace.
- Replace and Split modify or partition strings.

1.2 C# Structures (struct)

- Allow grouping fields of multiple data types (unlike arrays).
- Stored on the stack — efficient for small data.
- If a constructor is used, all fields must be initialised; otherwise default constructor zeros the fields.
- Value type — copies are real copies, but only when initialised.

1.3 Collection Classes

Collection	Use
ArrayList	Dynamic-size lists; insertion/removal/shuffling.
Dictionary	Key-value pairs; very fast lookup via hash table.
Queue	First-In-First-Out (FIFO).
Stack	Last-In-First-Out (LIFO).
Hashtable / SortedList	Hashtable allows any data type as keys; SortedList is a sorted Hashtable.

1.4 Embedded Review: MCU Architecture & Operation

- CPU = fetch–decode–execute cycle.
- Program Counter holds the address of the next instruction.
- ALU performs arithmetic and logic.
- Pipelining overlaps stages but is disrupted by branches.
- Von Neumann shares one bus; Harvard separates program/data buses for concurrent access.

1.5 Embedded Review: MCU Memory Types

- Flash: non-volatile firmware storage; finite write/erase endurance.
- SRAM: volatile run-time memory for variables, buffers, and stack; embedded C usually avoids the heap.
- EEPROM: non-volatile, byte-addressable for settings and calibration.

1.6 Embedded Review: Digital Communications

Protocol	Type	Key characteristics
UART / RS-232	Asynchronous serial	No shared clock; baud rate, start/stop framing; effective rate < baud.
SPI	Synchronous, full-duplex	SCK/MOSI/MISO/CS; master controls clock; high-speed continuous streaming.
I ² C	Two-wire synchronous bus	SCL/SDA; multi-device via addressing; pull-ups for open-drain.
CAN	Robust network	Multi-master; message-based arbitration; built-in error handling.

Differential signalling (RS-422/485) wins over single-ended for noise rejection and distance.

1.7 Embedded Review: Software Design and Estimation

- Architectures: simple polling loop vs event-driven (interrupts) vs RTOS (deterministic).
- Battery life: $I_{avg} = D \cdot I_{active} + (1 - D) \cdot I_{sleep}$ where D is the duty cycle.
- I_{avg} combined with battery capacity (mAh) gives expected lifetime.

1.8 Use Cases & Applications

Need	Pick	Reason
Ordered, indexable list with fast append	List<T>	Generic, dynamic; O(1) end-append, O(1) random access
Lookup by key	Dictionary<TKey,TValue>	Hash-table; O(1) average lookup
Stable, value-like grouping (point, color)	struct	Stack-allocated, no GC pressure for small data
Read-only sequence with reference semantics	IReadOnlyList<T>	Hides mutation, exposes only iteration/length
FIFO of work items	Queue<T>	Producer/consumer patterns
LIFO of state	Stack<T>	Undo stacks, parser scratch space

1.9 Worked Example: Word Counter

```
string text = File.ReadAllText("speech.txt");
var counts = new Dictionary<string, int>(StringComparer.OrdinalIgnoreCase);
foreach (var w in text.Split(new[] { ' ', '\n', '.', ',', '!', '?' },
    StringSplitOptions.RemoveEmptyEntries))
{
    counts[w] = counts.GetValueOrDefault(w) + 1;
}
foreach (var kv in counts.OrderByDescending(kv => kv.Value).Take(10))
    Console.WriteLine($"{kv.Key}: {kv.Value}");
```

1.10 Struct vs Class

Trait	struct	class
Type	Value	Reference
Default location	Stack (or inline)	Heap
Assignment / param	Copies the value	Copies the reference
Inheritance	No (only interfaces)	Yes
Default size guidance	$\leq \sim 16$ bytes	Larger / mutable / inheritance

1.11 Embedded Cross-Reference

If you only remember a few embedded facts to pair with this lecture's C# content:

- Pipelining and Harvard architecture (Lecture 1–2) explain why MCUs hit single-cycle execution.
- Memory hierarchy (Lecture 3, 13) is what your battery budget is competing for.
- $I_{avg} = D \cdot I_{active} + (1 - D) \cdot I_{sleep}$ is the formula every battery-powered design lives or dies by.
- I²C is open-drain with pull-ups; SPI is push-pull; UART has no shared clock — pick by pin count and rate.

Key Definitions

String (C#)

Specialised array of 16-bit UTF-8 characters.

Structure (struct)

Value type grouping multiple fields, stored on the stack.

Dictionary

Key-value collection backed by a hash table.

Embedded System

Hardware/software combo built for a specific function under constraints.

Microcontroller (MCU)

Single IC integrating CPU, memory, and peripherals.

Program Counter (PC)

Register holding the address of the next instruction.

ALU

Arithmetic Logic Unit; performs arithmetic/logical operations.

Pipelining

Overlapping CPU stages so multiple instructions are in flight.

Harvard Architecture

Program and data on separate buses for concurrent access.

SRAM

Volatile data memory for variables/buffers/stack.

EEPROM

Byte-addressable non-volatile memory for settings/calibration.

UART

Asynchronous serial protocol with baud-rate framing, no shared clock.

I²C

Two-wire synchronous bus with pull-ups; multi-device by address.

Differential Signalling

Two-wire complementary signalling rejecting common-mode noise.

Linear Regulator

Simple low-noise regulator that wastes power as heat.

Generic Collection

Strongly-typed collection (List<T>, Dictionary<K,V>) avoiding boxing/unboxing of value types.

IEnumerable<T>

Interface for read-only forward iteration; the basis for foreach and LINQ.

LINQ

Language-Integrated Query; chainable operators (Where, Select, OrderBy) on collections and other sources.

StringComparer

Helper that controls how strings are compared (case sensitivity, culture).

Sources

- Lecture 20.pdf